SPE-227736-MS

A Novel Water-Soluble Compounds Extraction Technique for Geochemical Surface Soil Exploration

Makar Silaev, Vyacheslav Grokhovsky, and Andrey Kovalskii, Aramco Innovations LLC, Moscow, Russia; Peter Birkle, Geology Technology Division at EXPEC ARC, Dhahran, Kingdom of Saudi Arabia

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

Soil salinization is a major environmental process in arid and semi-arid regions, whereby more than 37% of salinized soils are located in deserts. Traditionally considered a constraint for agriculture and land productivity, salinization also offers a unique opportunity for geochemical exploration. Water-soluble forms of elements in soils reflect changes of environmental settings and play a significant role in soil salinization. The accumulation of specific elements in shallow soils is utilized in the present study as indicator for their secondary migration and dispersion from deeper-situated strata. In particular, elements from Groups I and II of the periodic table (e.g., Na, K, Ca, Sr, and Li) display high mobility under saline and alkaline conditions, often accumulating in shallow soil layers due to capillary rise and evaporation. An efficient and rapid water-based extraction technique was developed with high-accuracy ICP-OES instrumentation to identify anomalies of such water-soluble forms of elements in soil samples. The presented methodology was applied to 36 soil samples—including 32 aridisols and 4 sabkhas—from geochemically diverse landscapes of the Arabian Peninsula to identify geochemical anomalies linked to deeper sources of strategic elements. Significant anomalies in the abundance of Li, Na, K, and Be were detected for specific samples with an excess of up to 40 times relative to the median abundance. Soils with detected anomalous concentrations of strategic elements can serve as alternative source for mining purposes. The developed technique provides a faster and more accurate analytical procedure of extensive sample volumes for the mapping and identification of enhanced elemental accumulation in soil.

Introduction

Soil salinization is a widespread process in arid and semi-arid regions, affecting more than 20% of the world's topsoil, with particularly high concentrations in deserts (37%) and steppe ecosystems (27%) (Bashour & Sayegh, 2007; Qadir et al., 2014: Eswar et al., 2021). The salts that accumulate in surface and near-surface soils act as carriers for mobile elements—particularly those of economic interest—and thus provide indirect evidence of deep-seated geochemical processes (Corwin & Lesch, 2005; Zhang & Lalor, 2002; Mishra et al., 2023). The composition of soluble salts in soils can reflect geochemical processes occurring at depth and serve as indicators of element migration and dispersion pathways (Corwin & Lesch, 2005).

The geochemical behavior of Group I and II elements of the periodic table, such as lithium (Li), sodium (Na), potassium (K), calcium (Ca), magnesium (Mg), strontium (Sr), and beryllium (Be), is especially relevant in this context. These elements exhibit high solubility and mobility under saline-alkaline conditions, often being transported from deeper strata to surface soils through capillary rise, groundwater discharge, or evaporative enrichment (Kabata-Pendias & Mukherjee, 2007; Kelley et al., 2003). Consequently, the analysis of water-soluble compounds in surface soils has emerged as a powerful tool for early-stage mineral exploration, particularly for strategic elements like Li and Sr that are increasingly critical in energy storage technologies.

However, traditional water-extraction protocols for geochemical soil analysis often fall short in detecting trace-level element anomalies, particularly in highly heterogeneous substrates like aridisols and sabkhas (Bashour & Sayegh, 2007). Sabkha soils, formed through episodic seawater intrusion and evaporation, are especially dynamic in their geochemical composition. These soils exhibit frequent changes in moisture content, salinity, and ion exchange capacity, making them both a challenge and an opportunity for geochemical studies (Khan et al., 2006; Chenchouni, 2017). While aridisols are typically more stable, they often show signs of carbonate accumulation and salt crust formation, offering additional layers of geochemical complexity (Khan et al., 2006; Arifuzzaman et al., 2016).

In response to these challenges, this study introduces a refined aqueous extraction protocol aimed at improving the recovery and detection of mobile elements from surface soils in arid environments. However, traditional protocols often lack the sensitivity required for detecting low-abundance strategic elements. The presented methodology was applied to 36 soil samples—including 32 aridisols and 4 sabkhas—from geochemically diverse landscapes of the Arabian Peninsula. By analyzing samples under both air-dry and wet conditions, and normalizing the results to average aridisol values, the study provides a nuanced view of elemental mobility, anomaly detection, and the role of soil moisture in modulating geochemical signals. This approach advances both the analytical precision and interpretive reliability of surface geochemical data, contributing to improved techniques for strategic mineral prospecting and environmental monitoring in saline and arid regions.

Material and methods

Materials

Soil samples were collected from eight geochemically distinct sites across the arid landscapes of Saudi Arabia. From each site, four surface soil samples were taken at spatial intervals of 10–15 meters, yielding a total of 32 aridisol samples. These soils exhibited a sandy loam texture, low organic matter, negligible vegetation, and poor aggregation, which are typical attributes of aridisol profiles in desert environments (Eswar et al, 2021; Bashour & Sayegh, 2007; Sharma & Singh, 2017). Carbonate concretions were observed in some samples, suggesting post-depositional carbonate accumulation or aeolian reworking under low-leaching conditions.

Samples were received in an air-dry state and prepared for chemical analysis. In parallel, four sabkha soil samples—representative of saline flats influenced by episodic seawater intrusion and rising groundwater—were collected in a wet state. These soils were structurally more coherent and showed extensive salt crystal formation and secondary carbonates, characteristic of saline environments undergoing cyclic evaporation and capillary salt migration (Bashour & Sayegh, 2007; Arifuzzaman et al., 2016). These samples were analyzed both as-is (wet) and after air-drying, to assess how moisture content influences the solubility of mobile metal ions.

Methods

Soil extraction and analysis. A modified aqueous extraction protocol was developed and applied to enhance the recovery of Group I of the periodic table (Li, Na, K) and Group II (Ca, Sr, Be) elements, which are particularly relevant to geochemical exploration due to their high mobility in saline-alkaline environments (Kelley et al., 2003; Bashour & Sayegh, 2007). This protocol aimed to improve the detection sensitivity of mobile and water-soluble metal ions in arid and saline soil samples (Fig. 1). A total of 36 soil samples (32 aridisols and 4 sabkhas) were processed. From each sample, 50 grams were sub-sampled using the quartering method, ensuring sample homogeneity and representativity. The samples were oven-dried at 35 °C to attain an air-dry state - a temperature chosen to preserve volatile compounds and moisture-sensitive solutes.

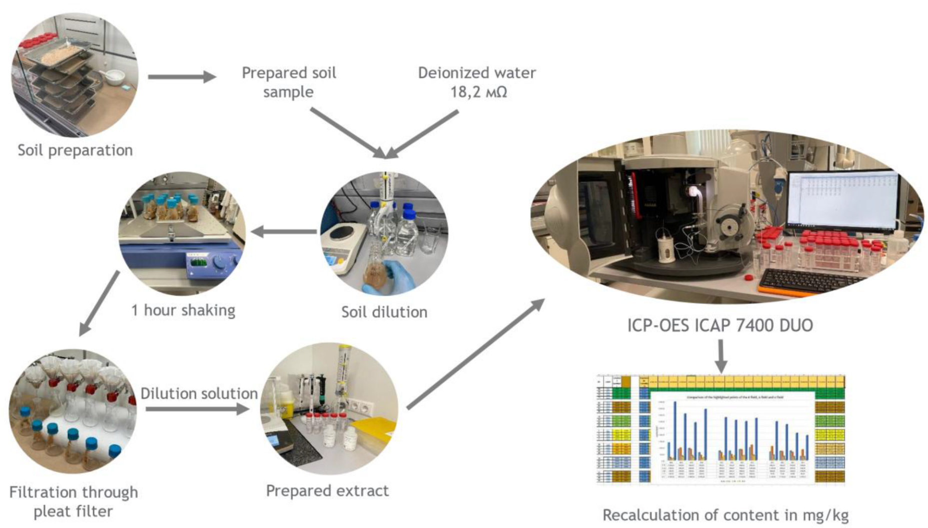

Figure 1—Workflow diagram illustrating the modified aqueous extraction protocol used for the analysis of mobile elements in aridisol samples. The process begins with air-dried soil preparation, followed by dilution with ultra-pure deionized water (18.2 MΩ). After 1 hour of orbital shaking, the solution is filtered through a pleated filter. The filtered extracts are then diluted and analyzed using ICP-OES. The final step involves recalculating elemental concentrations in mg/kg based on the dry weight of soil samples.

Dried samples were then ground in a porcelain mortar and sieved through a 1 mm mesh to break aggregates and increase effective sorption surface area, thus enhancing extraction efficiency. Precisely weighed portions of the processed soil were transferred into acid-washed conical flasks, where ultra-pure water (resistance: 18.2 MΩ·cm, Sartorius Arium® Mini system) was added in a 1:10 soil-to-water ratio (w/v). The use of ultra-pure water is critical for minimizing background ionic interference and improving the accuracy of low-concentration element detection.

The flasks were sealed and shaken for one hour on an IKA orbital shaker at consistent speed and temperature to ensure homogeneous solute diffusion into the aqueous phase. The resulting extract was passed through pleated filter paper to remove particulates and colloidal matter, followed by a 10-fold dilution by mass to ensure compatibility with downstream inductively coupled plasma optical emission spectrometry (ICP-OES) calibration ranges. The elemental concentrations of the filtered extracts were analyzed using ICP-OES with dual plasma view configuration. This configuration provides simultaneous axial and radial viewing, increasing sensitivity for trace and major cations (Zhang & Lalor, 2002; Corwin & Yemoto, 2020). Analytical calibration was performed using certified multi-element standards (Inorganic Ventures®) to ensure traceability and reproducibility across samples. Quality assurance included procedural blanks, spiked duplicates, and analysis of certified reference materials (CRMs) to confirm instrumental precision and matrix recovery.

Soil hydroscopic humidity. To ensure analytical accuracy and enable meaningful comparison across soil types, the hygroscopic moisture content of all soil samples was determined. A subsample of air-dry soil was placed in a vacuum drying oven and subjected to controlled drying at 105 °C under a reduced pressure of 650 millibars. This method enables the complete removal of sorbed water, including capillary and hygroscopic moisture, resulting in a true oven-dry mass reference state (Dane & Topp, 2020). Determining elemental concentrations on an absolutely dry mass basis is critical in geochemical analysis, as it eliminates variability introduced by differing residual moisture levels between samples.

For sabkha soils, which were received in a wet state, it was noted that transitioning to air-dry conditions might cause evaporative loss of mobile metal ions, particularly sodium, lithium, and potassium. Therefore, the initial wet mass and residual moisture content of each sabkha sample were carefully measured. Moisture loss was then factored in as a correction coefficient in subsequent calculations of metal concentrations, ensuring consistency between wet and dry sample datasets. This correction procedure allows the comparison of ICP-OES results on a standardized dry weight basis, thus enhancing the reliability of anomaly detection in saline environments (Hammer et al., 2011).

Results and discussion

Aridisols element concentration

The following elements were determined in 32 aridisol samples using ICP-OES at their respective wavelengths: Li – 670.7 nm, Na – 589.5 nm, K – 766.4 nm, Be – 313.0 nm, Mg – 279.5 nm, Ca – 317.9 nm, Sr – 407.7 nm, Ba – 455.4 nm. These elements are among the most mobile cations in saline-alkaline environments, often used as geochemical tracers for mineral weathering, groundwater influence, and elemental migration (Kabata-Pendias & Mukherjee, 2007; Alloway, 2013). To sanitize and standardize the data for more accurate interpretation, all elemental concentrations were converted into percentages of their average values. This allows direct comparison of anomaly magnitudes across elements with very different base concentrations and variance scales. Such normalization is crucial for detecting relative enrichment trends in heterogeneous soils like aridisols (Kabata-Pendias & Mukherjee, 2007).

The analytical results revealed considerable variability in element concentrations among the aridisol samples, reflecting both local geochemical heterogeneity and differential mobility of cations under arid conditions (Kelley et al., 2003). Summary statistics for all elements are presented in the form of a boxplot (Fig. 2).

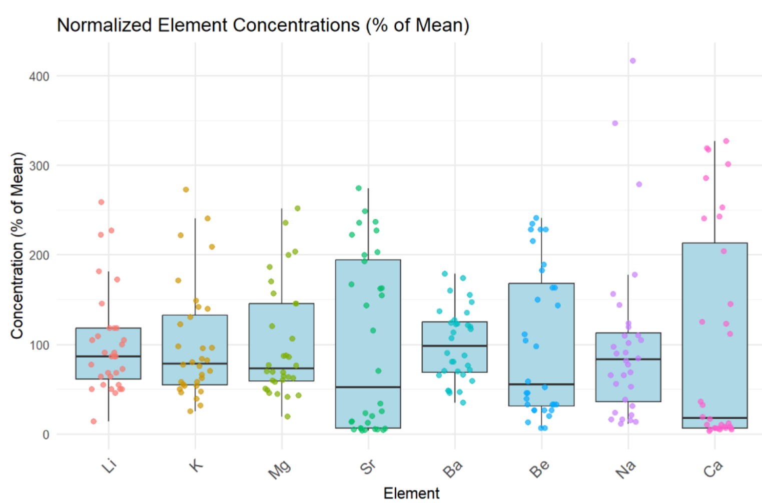

Figure 2—Boxplot of the dependence of the normalized concentration of elements with individual observations. Sr and Ca show the most variable results.

Element-Specific Observations

Each element's concentration was expressed:

Lithium (Li). Mean-normalized values ranged from 13.7% to 258%. Observed strong enrichment of 222% and 258% suggests geogenic input or deep-source migration of Li. Li anomalies above 200% of the mean are significant and match those seen in strategic element exploration (Kelley et al., 2003). The high Max/Min and Max/Median ratios also indicate the potential secondary redistribution of Li through capillary action or groundwater discharge.

Sodium (Na). Ranged from 11.1% to 416%. Elevated Na corresponds to field-observed surface efflorescence, which is consistent with previous studies on secondary salt crust formation in arid soils (Bashour & Sayegh, 2007).

Potassium (K). K ranged from 24.9% to 272.4% of average. According to (Alloway, 2013), K is often retained in clay minerals but may be solubilized under high ionic strength conditions. Notable are zones of weathering product accumulation, often associated with feldspar breakdown or saline groundwater transport (Alloway, 2013).

Beryllium (Be). Be displayed extreme outliers: AC-5 (188%), KE-3 (246%), and KE-4 (241%). These enrichments are possibly due to weathering of aluminosilicate minerals. Be mobility is limited, but in saline soils, complexation or electrostatic repulsion from colloids may influence its release (Bashour & Sayegh, 2007).

Magnesium (Mg). Mg anomalies peaked at two points (251% and 235%), with background-normalized levels typically between 40% and 90%. The consistency between Mg and Ca hotspots suggests carbonate mineral dissolution and reprecipitation (Bashour & Sayegh, 2007).

Calcium (Ca). Ca ranged from 0.5% to 327% of the mean. Some points showed extreme Ca enrichment, validating laboratory observations of carbonate-rich surface horizons typical of aridisols under strong evaporative regimes (Bashour & Sayegh, 2007) and may relate to gypsum dissolution/re-precipitation processes (Corwin & Yemoto, 2020).

Strontium (Sr). Highest anomalies: 274%, 221%, and 222%. Normalized Sr values above 200% of the mean are indicators of Ca-rich evaporites, since Sr substitutes for Ca in carbonates and gypsum (Corwin & Yemoto, 2020).

Barium (Ba). Ba showed lower relative variability; most values clustered around 80–140%. Point with concentration 179% was the most enriched, likely due to clay-bound Ba or residual feldspar weathering (Alloway, 2013).

Correlation between elements

The Pearson correlation matrix for the normalized elemental data provides clear insights into the geochemical processes affecting aridisol soils (Fig. 3). The most prominent trend is the strong positive association between lithium and magnesium ($r = 0.84$) and lithium and potassium ($r = 0.83$). These relationships suggest that Li, Mg, and K are mobilized together, likely due to weathering of silicate minerals and transport via saline groundwater and evaporative processes (Alloway, 2013). Such co-migration is typical in arid soils, where soluble ions accumulate near the surface.

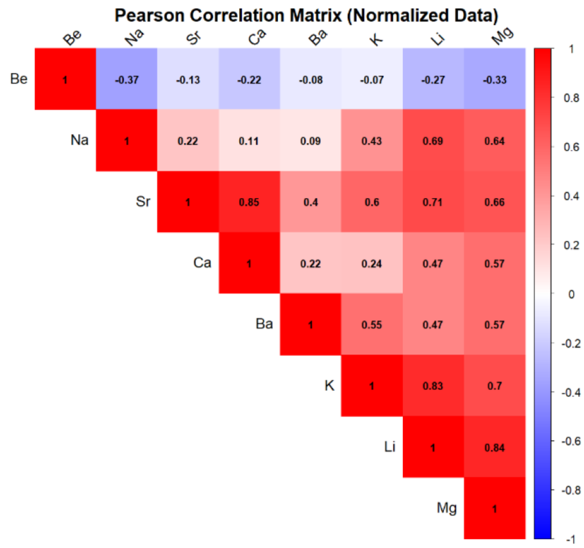

Figure 3—Pearson correlation matrix for normalized elemental concentrations in 32 aridisol samples. Strong positive correlations are observed between Ca–Sr (r = 0.85), Li–Mg (r = 0.84), and Li–K (r = 0.83), indicating co-migration under saline-alkaline conditions. Beryllium shows weak to negative correlations with most elements, suggesting distinct geochemical behavior and lower mobility in aridisol systems.

A similarly strong correlation is observed between calcium and strontium ($r = 0.85$), reflecting their geochemical affinity in carbonate and gypsum minerals. The substitution of Sr for Ca in these matrices is common under arid, evaporative conditions, where secondary precipitation dominates soil mineral formation (Kelley et al., 2003).

Sodium correlates well with lithium ($r = 0.69$) and magnesium ($r = 0.64$), reinforcing its role as an indicator of salinity and capillary salt movement. The presence of Na-rich anomalies aligns with field evidence of salt efflorescence and shallow water table effects in aridisol profiles (Corwin & Yemoto, 2020). In contrast, beryllium shows weak or negative correlations with most elements, including Na ($r = –0.37$) and Mg ($r = –0.33$). This suggests that Be is less mobile, likely retained in resistant silicate minerals or adsorbed onto oxide surfaces, consistent with its geochemical immobility in arid soils. These correlation patterns reveal two distinct element groups: a mobile cluster (Li, Na, K, Mg, Ca, Sr) that reflects evaporative concentration and hydrological transport, and a stable group (Be, partially Ba) tied more to primary mineralogy than to surface mobility. Understanding these associations is essential for identifying zones of strategic element accumulation and for refining soil-based geochemical exploration in arid environments.

Sabkha soils element concentration

Four sabkha soil samples were analyzed using Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES) under both air-dry and wet conditions to determine the concentrations of lithium (Li, 670.7 nm), potassium (K, 766.4 nm), magnesium (Mg, 279.5 nm), and calcium (Ca, 317.9 nm). To enable meaningful comparison with aridisol data, all concentrations were standardized as percentages relative to the average values observed in aridisol samples.

Figure 4 highlights the stark contrast in moisture content between the two sabkha sample conditions. While air-dry samples show near-zero moisture values, wet samples exhibit substantially higher humidity levels, ranging between 10% and 17%. This moisture discrepancy directly influences the solubility and mobility of water-soluble elements, thereby affecting their measured concentrations.

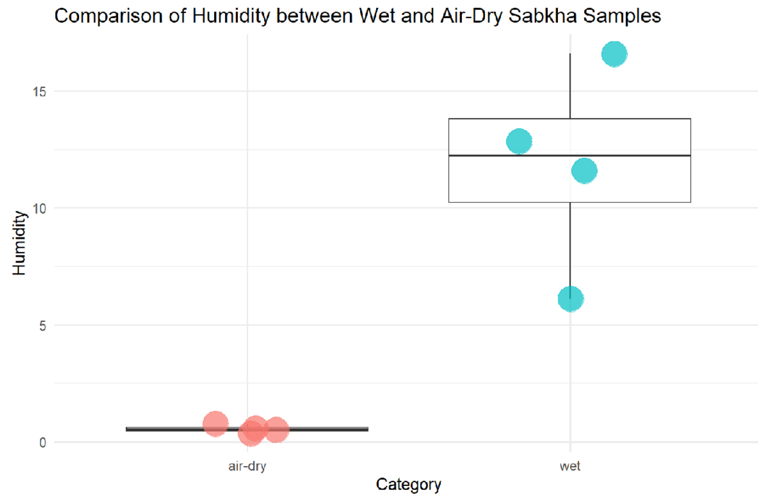

Figure 4—Comparison of humidity content between wet and air-dry sabkha soil samples. Wet samples exhibit significantly higher moisture values (ranging from 6% to 16%) relative to air-dry samples, which remain close to 0%, confirming successful differentiation between sample states.

Figure 5 presents normalized elemental concentrations for Li, K, Mg, and Ca in both air-dry and wet sabkha samples. The results reveal clear trends: potassium (K) exhibits the highest degree of variability and reaches its maximum concentration in the wet state, indicating its pronounced sensitivity to moisture-induced mobility. Lithium displays moderate increases in concentration from dry to wet conditions, while calcium and magnesium remain relatively stable and low in both states, suggesting limited solubility or stronger retention in the soil matrix. Also, lithium concentration reaches 20 times higher in sabkha soils compared to the aridisol average.

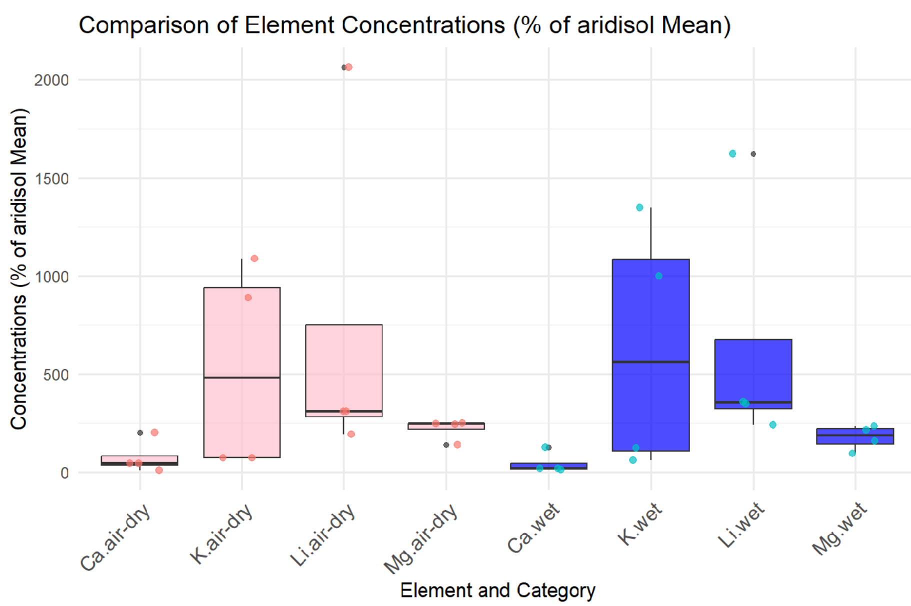

Figure 5—Elemental concentrations (Li, K, Mg, Ca) expressed as a percentage of aridisol mean under air-dry and wet conditions. Analyses performed using ICP-OES at specific wavelengths: Li – 670.7 nm, K – 766.4 nm, Mg – 279.5 nm, and Ca – 317.9 nm. Data standardized to facilitate comparative interpretation.

These findings underscore the critical role of soil moisture in modulating the mobility and bioavailability of mobile cations in saline environments. In particular, the sabkha system appears to respond dynamically to hydrological changes, with moisture-enhanced transport of certain elements, especially potassium, relative to aridisol baselines. When compared to aridisols, sabkha soils exhibit not only higher concentrations of mobile elements but also greater sensitivity to moisture content. This contrast emphasizes the importance of soil type and moisture regime in interpreting geochemical data for exploration or environmental assessment. In sabkhas, short-term hydrological changes may yield significant geochemical responses, particularly for elements like potassium and lithium that are both soluble and strategically important. From an exploration perspective, the sabkha's enhanced elemental signatures, particularly under wet conditions, could indicate secondary enrichment zones and act as surface indicators of deeper mineralization.

Conclusion

This study presents a modified aqueous extraction technique tailored for assessing mobile metal concentrations in surface soils from arid and saline environments. The method was successfully applied to both aridisols and sabkhas from the Arabian Peninsula, with elemental analyses performed using high-precision ICP-OES instrumentation. The results reveal substantial spatial and compositional variability in the distribution of Group I and II elements of the periodic table, including notable anomalies in lithium, potassium, magnesium, calcium, and strontium concentrations. Data normalization and comparison between air-dry and wet sabkha samples further demonstrate the significant impact of moisture content on elemental solubility and detectability, particularly for potassium and magnesium.

The proposed technique enables rapid, cost-effective screening for geochemical anomalies in surface soils and offers a promising tool for early-stage geochemical mapping and resource exploration in drylands. In a broader context, this approach enhances our understanding of element mobility under saline-alkaline conditions, aiding both environmental monitoring and strategic element prospecting in arid landscapes.

References

1. Alloway, B. J. (2013). Sources of heavy metals and metalloids in soils. *Heavy metals in soils: trace metals and metalloids in soils and their bioavailability*, 11–50.
2. Arifuzzaman, M.D., Habib, M. A., Al-Turki, M. K., Khan, M. I., & Ali, M. M. (2016). Improvement and characterization of sabkha soil of Saudi Arabia: A review. *Jurnal Teknologi (Sciences & Engineering)*, **78**(6).
3. Bashour, I. I., & Sayegh, A. H. (2007). Methods of analysis for soils of arid and semiarid regions (p. 119). Rome, Italy: Food and agriculture organization of the United Nations.
4. Chenchouni, H. (2017). Edaphic factors controlling the distribution of inland halophytes in an ephemeral salt lake "Sabkha ecosystem" at North African semi-arid lands. *Science of the Total Environment*, **575**, 660–671. https://doi.org/10.1016/j.scitotenv.2016.09.071
5. Corwin, D. L., & Lesch, S. M. (2005). Characterizing soil spatial variability with apparent soil electrical conductivity: I. Survey protocols. *Computers and electronics in agriculture*, **46**(1-3), 103–133. https://doi.org/10.1016/j.compag.2004.11.003
6. Corwin, D. L., & Yemoto, K. (2020). Salinity: Electrical conductivity and total dissolved solids. *Soil science society of America journal*, **84**(5), 1442–1461. https://doi.org/10.1002/saj2.20154
7. Dane, J. H., & Topp, C. G. (Eds.). (2020). Methods of soil analysis, Part 4: Physical methods. John Wiley & Sons.
8. Eswar, D., Karuppusamy, R., & Chellamuthu, S. (2021). Drivers of soil salinity and their correlation with climate change. *Current Opinion in Environmental Sustainability*, **50**, 310–318.
9. Hammer, E. C., Nasr, H., Pallon, J., Olsson, P. A., & Wallander, H. (2011). Elemental composition of arbuscular mycorrhizal fungi at high salinity. *Mycorrhiza*, **21**, 117–129.
10. Kabata-Pendias, A., & Mukherjee, A. B. (2007). Humans. In: *Trace Elements from Soil to Human*. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-540-32714-1_7

11. Kelley, D. L., Hall, G. E., Closs, L. G., Hamilton, I. C., & McEwen, R. M. (2003). The use of partial extraction geochemistry for copper exploration in northern Chile. *Geochemistry: Exploration, Environment, Analysis*, **3**(1), 85–104. https://doi.org/10.1144/1467-787302-048
12. Khan, M. A., Barth, H. J., & Böer, B. (Eds.). (2006). Sabkha Ecosystems: Volume II: West and Central Asia (Vol. **2**). Springer Science & Business Media.
13. Mishra, A. K., Das, R., George Kerry, R., Biswal, B., Sinha, T., Sharma, S., … & Kumar, M. (2023). Promising management strategies to improve crop sustainability and to amend soil salinity. *Frontiers in Environmental Science*, **10**, 962581. https://doi.org/10.3389/fenvs.2022.962581
14. Qadir, M., Quillérou, E., Nangia, V., Murtaza, G., Singh, M., Thomas, R. J., … & Noble, A. D. (2014, November). Economics of salt-induced land degradation and restoration. In *Natural resources forum* (Vol. **38**, No. 4, pp. 282–295). https://doi.org/10.1111/1477-8947.12054
15. Sharma, D. K., & Singh, A. (2017). Current trends and emerging challenges in sustainable management of salt-affected soils: a critical appraisal. *Bioremediation of salt affected soils: an Indian perspective*, 1–40. https://doi.org/10.1007/978-3-319-48257-6_1
16. Zhang, C., & Lalor, G. (2002). Multivariate relationships and spatial distribution of geochemical features of soils in Jamaica. *Chemical Speciation & Bioavailability*, **14**(1-4), 57–65.